

RRC Polytech campuses are located on the lands of the Anishinaabeg, Ininiwak, Anishininwak, Dakota Oyate, and Denésuline, and the National Homeland of the Red River Métis.

Course Outline

Course Information

Course Code and Title:	COMP-5015 – Programming 3
Department/Program:	Creative Arts / Video Game Development – Programming
Total Hours:	90
Credit Hours:	6

COURSE DESCRIPTION:

This course is an introduction to algorithms and design patterns. Students will learn to develop efficient and elegant software algorithms for complex problems faced by video game programmers. Students will also explore several advanced object-oriented data types and algorithms available in a modern object-oriented programming language and game engine.

RECOGNITION OF PRIOR LEARNING (RPL):

Recognition of Prior Learning (RPL) refers to a set of processes that allow individuals to document, be assessed and gain recognition for their prior learning. The focus is on the learning rather than where or how the learning occurred. Knowledge, skills and abilities gained from life experiences may be formal ([transfer credit](#)), informal or non-formal. RRC Polytech evaluates and grants credit for qualifying prior learning that is equivalent to the learning outcomes for courses in a program. For more information about RPL at RRC Polytech, refer to [RPL](#) or [Policy A14 – RPL](#).

Contact Chris Brower at cbrower@rrc.ca for information regarding RPL processes and opportunities for this course. For general information and assistance with RPL, contact RRC Polytech's RPL Advisor at 204.632.3094 or rpladvisor@rrc.ca.

ACCESSIBILITY STATEMENT:

RRC Polytech is committed to providing persons with documented disabilities fair and equal access to educational programs, services and facilities. If you are a student with a disability* and require reasonable accommodations, you must connect with Student Accessibility Services (SAS) who will assist in developing and implementing your accommodation plan. Refer to the [Student Accessibility Services webpage](#) for information about SAS locations and how to [book an appointment](#). Students with disabilities are also encouraged to have a private discussion with their instructor(s) to facilitate greater understanding of their learning needs.

*"Disability" refers to a condition that limits a person's daily activities. Persons with disabilities may have long-term physical, mental, intellectual or sensory impacts which may hinder their participation on an equal basis with others. A disability includes those related to or caused by aging, an injury or other life events that may temporarily or permanently affect mobility, dexterity (use of hands), vision, hearing, communication, understanding or mental health.

ALTERNATE FORMATS:

This content is available in alternate formats. To request, please contact accessibility@rrc.ca.

ACADEMIC INTEGRITY:

Academic Integrity means acting with the values of honesty, trust, respect, responsibility, fairness and courage in learning, teaching and research to

ensure that the credentials granted by RRC Polytech accurately represent demonstrated knowledge, skills and abilities. All members of the RRC Polytech community are expected to demonstrate these values through RRC Polytech learning activities, relationships and commitments. Clear expectations will be communicated to students to promote positive academic practices in compliance with RRC Polytech's [Policy A17 – Academic Integrity](#). Contact academicintegrity@rrc.ca for additional information.

COURSE DELIVERY METHODS:

This course may be delivered in a classroom environment, online, or in a blended manner.

The following communication tools will be used in this course:

- Academic Email
- MS Teams (Chat Channels and Blended Lectures)
- Learn News Postings

Course format:

Students will work through problems to gain a working knowledge of concepts. Assignments and projects are employed to evaluate understanding of core concepts and competencies. Class time will be a mix of lecture-based instruction, lab time, and work periods.

EFFECTIVE DATE:

August 25, 2025

Instructor Information

Instructor's name: Kyle Geske
Email: kgeske@rrc.ca
Office phone: 204-949-8335

Office location: W302

Office hours: Office hours by appointment. Please email or message me using Teams.

Student Readiness

TECHNOLOGY AND EQUIPMENT READINESS:

Weekly online or blended delivery will be required. As such, you will need access to a webcam, microphone, and headset.

STUDENT COMMITMENTS AND CONTACT TIMES:

- 6 hours mandatory scheduled daytime attendance per week.
- Students are expected to commit time outside of regularly scheduled classes to work on their challenges and their course projects.
- Students in this course are expected to regularly check the course MS Teams channels and Learn News.
- Students in this course are expected to regularly check their academic email account or ensure it is forwarded to another checked account.
- Instructors will notify students at the beginning of the term of any course-specific communication methods.

TEXTBOOK(S):

There is no textbook for this course.

COURSE RESOURCES:

The course notes are available in Learn and [by following this link](#).

Student Learning

While participating in this course, students will:

- Evaluate and refine ideas through iterative processes and incorporate feedback from instructors, peers, and testers.
- Analyze technical, creative, and operational problems through critical thinking to identify issues and develop solutions.
- Develop effective teamwork skills through active listening, constructive discussion, and collaborative problem-solving.
- Communicate effectively across various modes such as written, verbal, and digital, with consideration of diverse contexts and audiences.
- Exhibit leadership qualities by guiding others, encouraging cooperation, and contributing to the shared program objectives.

LEARNING OUTCOMES:

By the end of this course students should be able to:

1. Write and compile procedural and object-oriented programs in C++.
2. Design C++ programs that make use of:
 - A. User defined functions and lambda expressions.
 - B. User defined types.
 - C. Standard library and Unreal Engine container classes.
 - D. Custom object hierarchies using classes, inheritance, and polymorphism.
 - E. Pointers and references.
3. Make use of 3rd party object-oriented graphics and sound libraries to solve game development problems.
4. Implement, extend, and troubleshoot complex object-oriented game engine scripts.
5. Develop game-specific sub-systems in an efficient and well-designed manner.
6. Write stable and debug-friendly OO programs using logging and error handling mechanism.
7. Verbally explain to their instructor, and peers, the algorithms and design decisions used in the software they have written.

INSTRUCTIONAL SCHEDULE, ASSESSMENTS AND DATES:

NOTE: The following dates are subject to change based on the needs of the students at the instructor's prerogative. Students will be notified ahead of time of any changes.

Module	Topic	Learning Outcome(s)	Assessment and Evaluation
1	C++ Software Setup, History of C++, Tour of Language / IDE		
2	Introduction to C++	1	A01 – Introduction to C++
3	Standard Library Containers	1, 2C, 7	A02 – Working with Collections
4	User Defined Types	1, 2A, 2B, 2C, 2D, 6, 7	A04 – Working with User Defined Types
5	Sharpening your Saw with Exercism	1, 2A, 2B, 2C, 2D, 6, 7	A05 – Exercism Challenge
6	Introduction to a C++ Graphics Library	1, 2A, 2B, 2C, 2D, 3, 6, 7	A03 – Porting Applications to C++
7	Pointers and Polymorphism	1, 2A, 2B, 2C, 2D, 2E, 3, 6, 7	A06 – Pointers and Polymorphism
8	C++ in Unreal Engine	1, 2A, 2B, 2C, 2D, 2E, 3, 4, 5, 6, 7	A07 – Intro to C++ in Unreal Engine A08 – C++ Project in Unreal Engine

LETTER GRADE DISTRIBUTION:

Letter	GPA	Percentage
A+	4.5	90 to 100%
A	4.0	80 to 89%
B+	3.5	75 to 79%
B	3.0	70 to 74%
C+	2.5	65 to 69%
C	2.0	60 to 64%
D	1.0	50 to 59%
F	0.0	0 - 49%

A grade of D is required to pass this course.

Course Policies

GENERAL ACADEMIC POLICIES:

It is the student's responsibility to be familiar with and adhere to the RRC Polytech Academic Policies. These Policies can be found in the RRC Polytech calendar or online under Academic Matters at rrc.ca/legal/policies.

LATE SUBMISSION POLICIES:

Late submissions of coursework will result in an immediate 20% penalty deducted from the possible assignment mark, and another 20% for each 24-hour period that the assignment is late.

Each student is given two virtual “extension coupons”. Each coupon can be used to extend the due date of any assignment by one day. These coupons can be used with separate assignments (each assignment is extended by one day) or both coupons can be applied to the same assignment (the assignment is extended by due days). You **must** reach out to your instructor, in writing (email or Team DM), **before the due date** to make use of these coupons.

SUPPLEMENTARY POLICIES:

GENERATIVE ARTIFICIAL INTELLIGENCE (GEN AI)

The use of GenAI for academic work must follow the program and course expectations, as identified in this section and must adhere to [Policy A17 – Academic Integrity](#).

Generative AI systems, such as ChatGPT and Midjourney, are becoming invaluable tools in our industry. However, relying on them to complete your

coursework can rob you of an opportunity to learn foundational skills crucial for your success in the game development industry.

In this course, your primary goal is to develop and strengthen your own skills and understanding. While there may be instances where experimenting with and learning about generative AI is encouraged, it is vital that your work remains your own.

Generative AI Policy Overview:

- **Personal Effort and Understanding:** All coursework should be a genuine reflection of your personal effort and understanding.
- **Prohibited Use:** No part of any assignment or project should be generated by an AI tool unless explicitly instructed otherwise by your instructor(s).
- **Instructor-Directed Use:** There may be specific assignments where you will be asked to experiment with and learn about generative AI technology. In these cases, follow the guidelines provided for those assignments.
- **Consequences:** Any submission found to contain AI-generated output will receive a zero. Repeated violations may result in further disciplinary action in accordance with RRC Polytech's Academic Integrity policy.

ATTENDANCE POLICY

Regular class attendance is crucial for your success in this course. Excessive late arrivals or missed classes can hinder your learning and disrupt the classroom environment.

If you miss class **three** or more times, a meeting with the department chair will be scheduled to discuss your attendance and how it affects your progress in the course.

DATE REVISED:

May 16, 2025

Mental Health and Well-being at RRC Polytech

Having good personal health and well-being will support your success in this program.

WE ENCOURAGE YOU TO:

- Recognize that stress is an expected part of being a college student.
- Rethink how you view difficulty. Being challenged is actually a part of learning and reaching success.
- Reflect on your role in taking care of yourself throughout the term. Do your best to balance your schoolwork and life demands.
- Reach out to your instructor, or other College supports at any time if something is affecting your academic performance. It's always best to reach out early and it's the responsible thing to do.

COLLEGE SUPPORTS READY AND WILLING TO ASSIST YOU:

- [Academic Success Centre](#)
- [Campus Well-Being](#)
- [Equity, Diversity and Inclusion Supports](#)
- [Health Services](#)
- [Indigenous Student Supports](#)
- [International Student Supports](#)
- [Library Services](#)
- [Student Accessibility Services](#)
- [Student Counselling Services](#)
- [United Way 211 community resource](#)

Authorization:

This course is authorized for use by:



Chris Brower

Chair, Digital Media Design and Game Development

August 25, 2025

Date

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Approved by Senior Academic Committee March 2025

Please retain this course outline for future educational and/or employment use.